YOONHYUCK WOO

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RESEARCH INTERESTS

- Natural Language Understanding, Knowledge Graphs, Large Language Models (LLMs), Knowledge-Grounded Reasoning, Retrieval-Augmented Generation (RAG), Human-AI Interaction
- Research goal aims to advance human-centered NLP applications, such as intelligent chatbots, by integrating structured knowledge representation with deep semantic understanding grounded in language model-based reasoning.

EDUCATION

Purdue University, West Lafayette, IN, USA **Aug. 2025 – Present**
Ph.D. in Computer and Information Technology Advisor: Dr. Julia T. Rayz

Purdue University, West Lafayette, IN, USA **Aug. 2023 – Aug. 2025**
Master of Science in Computer and Information Technology Advisor: Dr. Julia T. Rayz
Thesis: *Analyzing Knowledge Graph Reliability and Semantic Coherence Through Embedding Clustering*

Jeonbuk National University, Jeonju, Korea **Mar. 2017 - Feb. 2023**
Bachelor of Science in Industrial Engineering Advisor: Dr. Jaekyung Yang
Graduated with highest academic standing

PUBLICATION & PRESENTATION

Y. Woo, J Rayz, "Probing Knowledge Graph Reliability and Semantic Coherence with Language Models." Accepted to *Proceedings of the 39th International Florida Artificial Intelligence Research Society Conference (FLAIRS 2026)*.

D. Luo, I. Yang, J. Bae, and **Y. Woo**, "Research on Performance Metrics and Augmentation Methods in Lung Nodule Classification," *Applied Sciences*, vol. 14, no. 13, p. 5726, 2024.

D. Luo, J. Bae, **Y. Woo**, and J. Baek, "Improve Pulmonary Nodule Classification Performance Utilizing Proper Metrics and Augmentation Methods," in *Proceedings of 2022 Korean Institute of Industrial Engineers Autumn Conference*, Incheon, Korea, Nov. 2022.

Y. Woo, I. Choi, and J. Bae, "Automation of lung nodule diagnosis with Transformer segmentation model," *presented at the 2022 Korean Institute of Industrial Engineers Local Conference*, Gwangju, Jeollanam-do, Korea, Dec. 2, 2022

RESEARCH EXPERIENCE

IMLS-Funded Project, Purdue University Libraries & School of Information Studies **Jan 2026 – Present**
Graduate Research Assistant Advisor: Dr. Gang Shao
- Supported the development of a dual-path learning framework combining code-based and no-code AI tools, while contributing to the design of practical AI learning materials for AI-enabled library services, instruction, and research support.
- Designed and delivered an NLP workshop through Graduate & Postdoctoral Professional Development, Purdue University.

Applied Knowledge Representation and Natural Language Understanding Lab, *Purdue University* **Feb. 2025 - Aug. 2025**
Graduate Research Assistant Advisor: Dr. Julia T. Rayz
Project: Understanding Resident Responses to Thermostat Changes Using LLMs
- Developed a two-module LLM system to detect reasoning in resident responses and generate natural follow-up prompts.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline to answer resident queries using domain-specific documents.
- Performed clustering analysis on real resident responses to evaluate category alignment and optimize labeling structure.
- Analyzed embedding distances to assess semantic consistency across predefined and emergent categories.

Management Information System Lab, *Jeonbuk National University* **Aug. 2022 - Dec.2022**
Undergraduate Research Trainee Advisor: Dr. Joonsoo Bae
- Conducted research related to diagnosis of lung nodules by using deep learning

Cognitive Computing Lab, *Jeonbuk National University* **Jan. 2021 - Jul. 2022**
Undergraduate Research Trainee Advisor: Dr. Seung-Hoon Na
- Conducted NER in English and Korean, implemented Linear Regression and MLP models from scratch with NumPy, and extended experiments with the published model SPANNER model for Korean

TECHNICAL SKILLS

Languages: Python, C, R
ML / DL Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
Data & Scientific Computing: NumPy, pandas
LLM & AI Systems: RAG, LangChain
Platforms: Google Colab, Jupyter Notebook, Overleaf

PROJECTS

Conference Knowledge Synthesis Assistant **Feb. 2026 - Apr.2026**
Interactive Systems and Web Development, *Purdue University* Professor: Dr. Tianyi Li
- Engineered a knowledge graph system to synthesize insights across multiple research papers and user notes, improving literature

review efficiency

- Integrated LLM-based candidate generation with code-driven validation and grounding to improve reliability
- Implemented project-level KG construction with automated concept merging, cross-paper relationship detection, and interactive visualization (Django, D3.js)

Dynamic Context-Aware Chatbot using ConceptFlow **Mar. 2024 - Apr.2024**

Human-AI Interaction, *Purdue University* Professor: Dr. Ming, Yin

- Enhanced a chatbot by integrating published model ConceptFlow with advanced graph attention mechanisms and concept-based strategies, and developed a user-friendly interface to improve interaction and context management
- Propose a preprocessing function for new posts to facilitate real-time response generation within the ConceptFlow

LEADERSHIP & MILITARY SERVICE

Academic Mentoring Program | *Mentor*, Jeonbuk National University **Sep. 2022 – Dec. 2022**

- Led a mentoring program for three undergraduate students, managing regular sessions and mentoring fairs to support their academic and professional success.

Republic of Korea Army | *Sergeant,* Telecommunication Company 3rd Engineer Brigade, 3rd Corps **Apr. 2018 – Dec. 2019**

- Operated C4I systems as a Squad Leader, managing task allocations and mediating team conflicts.
- Twice awarded the Commendation of 3rd Engineering Brigade Commander for exemplary service.

AWARDS & SCHOLARSHIPS

- Study Abroad Scholarship, Jeonbuk National University ($9167.20) **Jul. 2023**
- President's Awards for Academic Excellence (Outstanding Graduate), Jeonbuk National University **Feb. 2023**
- Academic Scholarship, Jeonbuk National University **Aug. 2017-Feb. 2018/Aug. 2020-Feb. 2022**
- ABEEK (Accreditation Board for Engineering Education of Korea), 15th National Portfolio Competition, Bronze prize **Oct. 2021**